

Rules of Design<=>Specification engagement

Abstract

First things first: this paper is a process description, and the main audience is Working Group and Study Group chairs, their assistants, and WG21 expert members who do proposal reviews. This paper is not a tutorial for writing good proposals - although keen-eyed readers may find the description useful for producing high-quality proposals.

This paper explains the rules of engagement that have been in use between Evolution and Core, amends the rules with regards to Evolutionary specification review, and proposes adopting the same rules between Library Evolution and Library.

The amendment to the rules should change nothing for Core and Library (well, not for worse, at least), it's merely a specification-runthrough in EWG and LEWG before the material proceeds for technical/specification review. But as a formal addition, there is a step after the specification review to review available implementation/deployment experience; this review is to be done by both the Design and the Specification group. Whether the groups want to combine forces for that review or perform it separately is left up to their discretion, but both groups should perform that review, because they will be doing so from different viewpoints with a different emphasis.

The proposal ALSO proposes a new Tentatively Plenary state between specification review and plenary poll. This is proposed in order to give reviewers time to digest new proposals. Said state can be bypassed in urgent situations, but by default, it's proposed in order to allow for gathering more implementation/deployment experience, experiment feedback, and other feedback.

To repeat and rephrase, this proposal attempts to improve the following abilities:

- the ability of CWG/LWG to even start processing an incoming paper
- the ability of CWG/LWG to process an incoming paper without unnecessary overhead and waste of time
- the ability to give post-EWG/LEWG-discussion feedback on proposals
- the ability to review proposals after CWG/LWG review but before the plenary
- the ability to verify that a change in the standard is implementable
- the ability to verify that a change in the standard is deployable

The <=> in the title is not a spaceship, it's two overlapping arrows going into opposite directions.

Revision history

Highlight the most recent changes: ☐

Highlight the R3->R3 changes: ☐

Highlight the R1->R2 changes: ☐

Highlight the R0->R1 changes: ☐

- R4: This version, edited the remaining mentions of "Tentatively Ready" to say either "Tentatively CWG/LWG" or "Tentatively Plenary", like in the diagram. Also added a process bullet that tells not to remove design/rationale bits when proceeding to specification review
- R3: Clarifications for intended audience and how to solicit assistance in writing wording. Also clarify the conditions for bypassing Tentatively CWG/LWG/Plenary states, and clarify that Tentatively Plenary means the matter will be in a plenary straw poll in the next meeting.
- R2: New additions for
 1. a pre-plenary Tentatively Plenary stage both in the diagram and in the description,
 2. the post-design Tentatively CWG/LWG status is now in the diagram, and in the descriptionAlso changed from "wording" to "specification".
- R1: Added clarifications for
 1. what the qualifications of a wordsmith are,
 2. how to find wordsmiths, and
 3. how a WG can choose to do a small-group specification pre-review instead of relying on a single individual wordsmith to do it
- R0: Initial version

Further elaboration on the problems this proposal aims to tackle

Here's an alternative way to put it, in three parts:

1. Avoid wasting time in the various stage of the pipeline. Especially avoid wasting time in the stages that are our bottlenecks, aka the specification review.
2. Define what it requires to advance from one pipeline stage to the next.
3. Define checkpoints for going back from one pipeline stage to an earlier one, if necessary.

Here's why we propose to introduce new checkpoints for possibly going back to earlier pipeline stages, and the problem we're trying to tackle by doing so, as phrased by Peter Dimov:

Roughly and approximately, the problem that the committee can potentially inflict billions of dollars of damage on the C++ community by breaking code in a sufficiently problematic way. This requires a certain amount of due diligence.

Some remarks of the alleged downsides of this proposal

The following concerns might be raised about this proposal:

1. it introduces very stringent gatekeeping
2. it makes the process harder for new committee members

For the first, it should be pointed out that this proposal doesn't actually introduce anything new; we have done all of the things in it in an inconsistent fashion, sometimes doing it, sometimes not. This proposal changes what our defaults are, and allows bypassing those defaults if we make an Educated Decision to do so.

For the second, this proposal makes our pipeline stages officially-specified for everybody, both old and new committee members. And it gives new and old committee members a process specification that they can read, as opposed to relying on hearsay and never really knowing what to expect.

Finally, while stringent gatekeeping might seem like a nuisance to some, and a multi-stage process that requires multiple championing steps might be inconvenient for proposal authors who can't attend multiple meetings, all that must be secondary to the ultimate goal of all of this, which is a higher-quality International Standard. This is simply not negotiable.

Some historical background

The original problem

Way back when, not really important when, but a couple of years ago, we had three problematic issues with material that flowed from Evolution to Core:

- some material didn't have specification when it arrived on Core's plate.
- some material had specification that hadn't been pre-reviewed by a Core expert (in some cases this was because the specification had been written by people who over-appreciated their level of specification expertise).
- some material needed to be sent back to Evolution, and Evolution wasn't always prepared nor welcoming of such an event of material being sent back.
- similar issues occurred between LWG and LEWG, for all of those cases.

The solution

To alleviate these problems, the Chair of Evolution specified the following rules, although they weren't written down (a snag that this document aims to rectify):

- a proposal forwarded to Core must have wording; this is not negotiable, and Core shouldn't and doesn't entertain suggestions for fishing expeditions, they are busy.
- such wording must be either written by a recognized wordsmith or reviewed by one.
- if, for any reason, Core thinks they need a design clarification or have any other question or a challenge about the design, the rule is to send the material back to Evolution accompanied by a member of Core that can explain the problem.

Notably, regardless of whether it's

- a "please clarify", or
- a "please reconsider", or
- a "we think this stinks",

the rule is the same - send it back and worry about possible hurt egos later. Core is not a design group, but they have enough experience that they are allowed and invited to challenge designs that make no sense to them. And in any case, Core shouldn't try to resolve design issues in Core, that's Evolution's job and Evolutions responsibility, and at the very least Evolution should understand the Core-technical consequences of their decisions.

The rationale for these points shouldn't be hard to understand:

- Core (or Library) is supposed to review a technical specification and make sure it's worded suitably, accurately, and consistently, and wording-wise and otherwise fits into the language. Doing so requires having the actual wording that is proposed to be incorporated into the Working Draft.
 - Core (or Library) doesn't write wording.
 - Some members of Core (or Library) might be so kind to help write wording.
- Core's (or Library's) workload is hefty, and the time of the experts in that room is not used well if they are looking at a specification that is nowhere near ready to have a chance to survive the first 5 minutes of review before there is a request to go away and come back with proper wording. This is both about consistency and completeness - the wording should be roughly similar to the wording we already have terminology-wise, style-wise and otherwise, and it must have all the things that are part of the design.
 - Core (or Library) doesn't write wording.
 - Some members of Core (or Library) might be so kind to help write wording.

The process

Once these issues were recognized and the alleviation for them agreed on, we established the following process for Design->Specification:

- Evolution reviews a proposal, and decides they are happy with it and want to forward it to Core.
- The EWGChair queries/verifies the existence of the wording.
 - In case it doesn't exist, the instructions for the paper author are "find a wordsmith, produce wording, and report back to me".
 - In case it exists, the question is "has it been reviewed by a reputable wordsmith? If not, have it so reviewed and report back to me."
- Once EWGChair is satisfied with the reputable wordsmith's report that the wording is ready for Core consumption, Core is made aware of incoming material and the proposal author is greenlighted to proceed to Core review at Core's scheduling pleasure.

And, as mentioned earlier, the following process for Design<-Specification:

- Core reviews a proposal, and discovers a design issue.
- The CWGChair notifies EWGChair that something needs to come back, and designates a Core champion to explain the issue in Evolution.
- EWGChair schedules such a discussion with elevated priority, and we iterate again, with the Design<->Specification rules as before.

The New Rules

These rules apply to both EWG->CWG and LEWG->LWG. Furthermore, there is now a new Tentatively Plenary step between specification review and a plenary poll.

Rather than the chair of a design group making sure that wording exists and is ready for the technical/specification review group's consumption, that responsibility now lies with the whole design group. That is, once the proposal author has made sure their wording is either written by a wordsmith or reviewed by one, they approach the

design group (with the wordsmith) and give a presentation about the wording, illustrating how it implements the design, and report whether any oddities or questions surfaced. If such oddities or questions did surface, the design group needs to discuss them and possibly clarify their design.

This rule change is made for a couple of reasons:

- the design groups must understand the technical consequences of their designs
- the design groups must double-check that the design is accurately implemented by the specification, instead of saying very late in the process that "that's not what we meant" or "that's against the design intent".

"What's a 'wordsmith'? Who qualifies?"

A wordsmith is, by the definition for the purposes of this process, "a person who can write or help write CWG/LWG-consumable wording". It's ultimately the decision of the chairs of EWG/LEWG to decide what sort of wording they're willing to accept as wording incoming into their specification review (which might be less strict than the specification-group review), and those groups should have confidence that what they forward is written and championed by a person that won't be wasting Core's or Library's time.

CWG and LWG can also, if they so choose, arrange for a small group of their members to do this specification-work. Note, however, that such arranging must not consume group time, and is not the responsibility of the chair of the specification review group.

The solicitation for wording assistance is highly recommended to have roughly the following priority order:

1. The members of EWG/LEWG should be queried for volunteers to help with wording when the initial design review is done.
2. Well-known wordsmiths can be asked for help by more seasoned EWG/LEWG members, which is a slight variation of the above.
3. Wording assistance requests can be made on the reflector.

It is not appropriate to spend CWG's or LWG's time in the quest for a wordsmith, as a group.

Further explanation of the new process

So, to sum it up, let's bulletize the Design->Specification process:

1. The design group, be it LEWG or EWG, decides after reviewing a proposal that it should go forward to LWG or CWG.
2. In all the subsequent steps, both the wordsmith and the proposal author/champion should be available and present. That includes both the Design group's wording review and the Specification group's wording review.
3. If the proposal has wording, the design group must run through it and verify that the design is wholly and accurately implemented by the wording, and that there are neither omissions nor additions that the design doesn't cover.
4. If the proposal does not have wording, the design group must not greenlight the proposal to go forward; they must instead instruct the proposal to come back with the wording.
5. When the proposal has new wording not yet reviewed in the second bullet, go back to the third bullet. Repeat as many times as necessary.
6. Once the run-through is complete, and the design group is confident that the wording implements the design wholly and accurately, the design group sends the material onwards, forwards.
 - By default, that means
 - a. first parking it for further design feedback,
 - b. possibly reiterating the design based on such feedback, if any,
 - c. and then making it Tentatively CWG/LWG
7. When the material is forwarded, all rationale, design discussion, motivation, and other design-level information Shall Be Kept In It. The design group shall make no request to remove it, and the design group shall make no request to produce a "wording-only" paper. The specification review group will need that information to verify that the specification implements the design intent. Future readers investigating the paper trail will also need this information.

Let's also bulletize the Design<-Specification process, and also the steps between Specification->Plenary:

1. The specification/technical review group reviews a proposal and discovers a design-level addition/omission/bug/issue/snag, or a technical problem (an implementability problem, a consistency problem, or some other problem).
2. The chair of the specification/technical review group makes sure the problem is minuted, sends a heads-up to the chair of the design group, and designates a champion that can explain the problem to the design group.
3. The design group reviews the matter, and resolves it according to the Design->Specification process.
4. The specification/technical review group continues progress on the proposal.
5. Once the specification review is complete, both the Specification group and the Design group look at the available implementation/deployment experience of the proposal.
6. If that experience is deemed sufficient, the proposal is made Tentatively Plenary, which means it's ready for a plenary poll.
7. If there's feedback that necessitates looking again at either the design or the specification, the proposal is sent back to those stages.
8. Otherwise, proceed to the plenary poll *in the next meeting*.
9. Bypassing the Tentatively CWG/LWG status after design review or the Tentatively Plenary status after specification review is allowed by the process, but
 - such exceptional bypasses shall happen only after an explicit decision made by the design group (in the case of bypassing Tentatively CWG/LWG) or by both the design group and the specification group (in the case of bypassing Tentatively Plenary)
 - such decisions shall be made after a minuted discussion.

Or in a diagram:

EWG/LEWG guidance phase



EWG/LEWG design decision phase



Specification review phase



Finishing touches phase



